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ORIGINAL ARTICLE

Rhinitis, Sinusitis and Upper Airway Disease

Forecasting daily total pollen concentrations on a global scale

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Abstract

Background: There is evidence that global anthropogenic climate change may be impacting floral phenology and the temporal and spatial characteristics of aero-allergenic pollen. Given the extent of current and future climate uncertainty, there is a need to strengthen predictive pollen forecasts.

Methods: The study aims to use CatBoost (CB) and deep learning (DL) models for predicting the daily total pollen concentration up to 14 days in advance for 23 cities, covering all five continents. The model includes the projected environmental parameters, recent concentrations (1, 2 and 4 weeks), and the past environmental explanatory variables, and their future values.

Results: The best pollen forecasts include Mexico City (R^2(DL_7) ≈ .7), and Santiago (R^2(DL_7) ≈ .8) for the 7th forecast day, respectively; while the weakest pollen forecasts are made for Brisbane (R^2(DL_7) ≈ .4) and Seoul (R^2(DL_7) ≈ .1) for the 7th forecast day. The global order of the five most important environmental variables in determining the daily total pollen concentrations is, in decreasing order: the past daily total pollen concentration, future 2 m temperature, past 2 m temperature, past soil

For affiliations refer to page 2183.

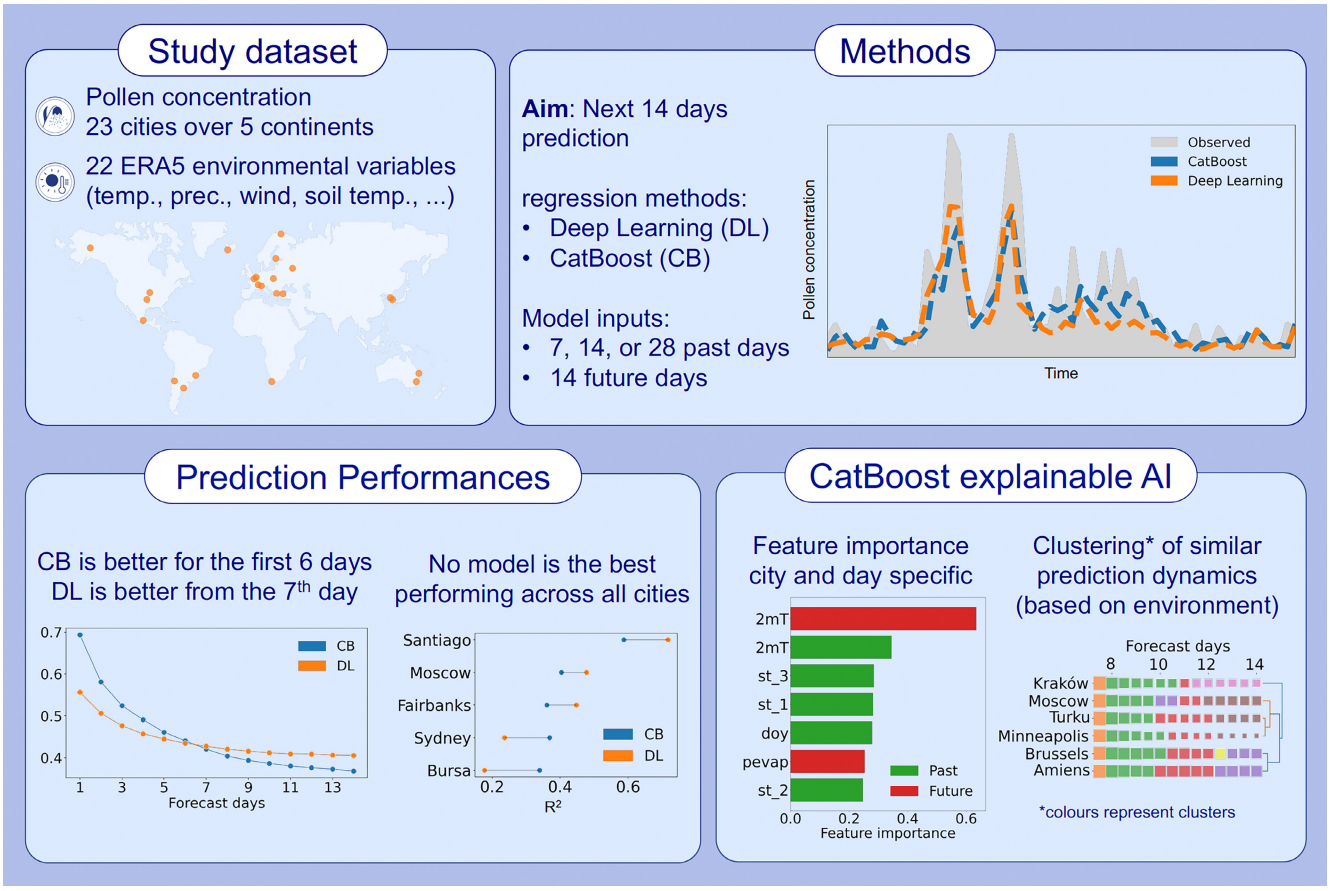
[Correction added on 17 July 2024, after first online publication: Affiliation 6 has been corrected in this version.]

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temperature in 28–100 cm depth, and past soil temperature in 0–7 cm depth. City-related clusters of the most similar distribution of feature importance values of the environmental variables only slightly change on consecutive forecast days for Caxias do Sul, Cape Town, Brisbane, and Mexico City, while they often change for Sydney, Santiago, and Busan.

Conclusions: This new knowledge of the ecological relationships of the most remarkable variables importance for pollen forecast models according to clusters, cities and forecast days is important for developing and improving the accuracy of airborne pollen forecasts.

KEYWORDS
allergy, artificial intelligence, environmental variables, feature importance cluster, pollen forecast



GRAPHICAL ABSTRACT
CatBoost is a preferable model for short-term forecasts, while Deep Learning is for longer ones, but there is no definite answer to what the better model is for every day or city. Past pollen trends are strong indicators of future pollen concentrations. CatBoost can be used to determine the importance of environmental variables in forecasting daily total pollen concentration. Abbreviations: 2mT, 2 m temperature; CB, CatBoost; DL, Deep Learning; DOY, day of the year; ERA5, the fifth generation ECMWF (European Centre for Medium-Range Weather Forecasts) atmospheric reanalysis dataset; pevap, potential evapotranspiration; st, soil temperature.

1 | INTRODUCTION

Pollen has been associated with allergic symptoms in sensitized individuals, having a direct negative impact on the overall quality of life of a substantial part of the population. Globally, the prevalence of allergenic pollen-induced asthma in different countries has been considered to range from 1% to 18% of the population.[1,2] Up to 80% of people with asthma also have allergies like hay fever, either seasonally or all year

long.[3] In 2016, it was estimated that more than 340 million people had asthma globally[4] with 420,000 deaths associated with asthma at the global level.[5] By 2025, the number of people with asthma may increase by an additional 100 million,[1] and asthma occurrence in urban populations is projected to increase from 45% to 59%.

Health implications other than rhinitis and asthma, have also been reported. A strong association was found between pollen concentrations and deaths due to cardiovascular diseases, Chronic Obstructive Pulmonary Diseases (COPD) and pneumonia.[6,7] In addition, airborne pollen might be an environmental risk factor for heart attack.[8] Interestingly, high pollen concentration was also found to exert a strong relation with stroke[9] and female suicide mortality.[10] Based on a comprehensive study, with in vitro, in vivo and human cohort studies, it was detected that coexposure to airborne pollen enhances susceptibility to respiratory viral infections, independent of pollen sensitivity.[11] In addition, Damialis et al.[12] found that higher airborne pollen concentrations correlate with increased SARS-CoV-2 infection rates.

As there is still no definitive cure, pollen allergy sufferers should avoid exposure to high concentrations of pollen allergens. To this end, pollen forecasting is an important tool that should be reliable and easily accessible.

Given the links between aero-allergenic pollen and weather, accurate and time-sensitive pollen forecasts would allow for preventive and prophylactic measures to help reduce allergic rhinitis, asthma, and other physiological consequences.

Computational Intelligence (CI) methods (1) can deal with the complexity of the release and dispersion of airborne pollen,[13] (2) are computationally efficient,[14] (3) can be easily integrated into operational use of the models,[15] and (4) provide substantially higher forecasting power compared to traditional regression models.[16] The main disadvantage of CI methods is their lower interpretability,[15,17] their limited effectiveness where smaller data is present (e.g., at the beginning and end of the pollen seasons),[18] and their sensitivity to the training data.[15,18]

In the literature, CI methods have been used for forecasting (a) daily pollen concentrations,[19,20] (b) pollen-induced symptoms,[21] and (c) risk level of pollen in the air[21,22] (Supplement, Section S1).

The aim of the study is (1) to develop and apply accurate and effective forecasting techniques to reliably predict daily total pollen concentrations (namely, the daily sum of airborne pollen for all taxa per cubic meter of air) on an expanded geographic scale, (2) to evaluate the overall order of importance of influencing variables in determining future pollen concentrations, and (3) to explore the physical-environmental associations for pollen responses as a function of the geographical location at a global level.

2 | MATERIALS AND METHODS

2.1 | Location and data description

We initiated a global survey for pollen data for all continents. Criteria included the availability of uninterrupted pollen concentration time series of at least 20 years. Among the responses, a total of 18 stations satisfied the selection criteria. However, to consider larger areas (especially in the southern hemisphere), we included stations from Caxias do Sul (Brazil), 8 years; Mexico City (Mexico), 13 years; Santiago (Chile), 7 years; Sydney (Australia), 12 years and Brisbane (Australia), 6 and 5 years in two parts. In total 23 stations were included (Figure 1; Supplement, Table S1), belonging to different climates (Supplement, Section S2.1.1). (Hereinafter, "S" in the numbering of figures and tables indicates that they are found in the Supplement). The name of the taxa, listed per station, and measurement period are included in Supplement I.

Our focus was on predicting daily pollen concentration without emphasizing the species, as, given the geographic diversity, equal species relevance was not possible. Therefore, we took into account the daily total pollen concentrations for every city (Supplement I).

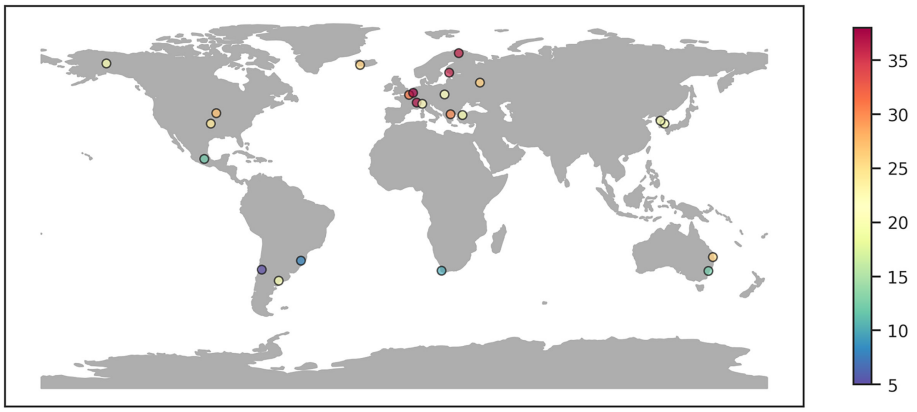


FIGURE 1 Geographical distribution/location of the aerobiological stations that provided data for the analysis. The colours refer to the number of years considered with the available daily pollen concentrations. [median: 22 years; min: 5 years, Santiago (Chile); max: 38 years, Brussels (Belgium)].

2.2 | Methods

We used gradient boosting (GB) with decision trees and deep learning (DL) models.[13,21,23–25] We implement GB with CatBoost (CB) library, which builds an ensemble of regression trees that reduces overfitting risk and improves accuracy.[26,27] The DL model follows an encoder-decoder architecture that makes use of 1d-convolutions and long short-term memory (LSTM) layers. By using LSTM layers the DL model outputs a 14-day pollen concentration projection. For GB instead, we train 14 models to predict the daily total pollen concentration for a 14-day projection (one model per day).

We compare the CB and DL models with the persistence model, which propagates the last daily total pollen concentration record up to 14 days ahead. We use the persistence model as a baseline model.[22]

Models are trained to forecast pollen concentration for the next 14 days, using as input the 22 environmental variables, the day of the year, the city location and the observed daily total pollen concentration. All variables with the exception of pollen concentration are used to model both the past and the future, while pollen concentration is present only for the past. We considered daily values of either 7, 14, or 28 days for the past context. We also trained the models without the input pollen concentration information, to assess the benefits of employing an autoregressive model. The environmental parameters considered in this study come from the ERA5-Land reanalysis, from which we draw both past and future values. In an operational setting, future values would result from environmental forecasting models.

To conduct the machine learning experiments, we divided the data of each station into training (50%), validation (25%), and test (25%) sets based on the year. For example, Tulsa (OK, USA) provided data for the years 1996–2019: 1996–2007 was used as a training set, 2008–2013 as a validation set, and 2014–2019 as a test set. This is a common practice adopted in machine learning.[28–30] This validation scheme aims to closely represent operational conditions, where observed climatic shifts are included in the available training data.

We report model performances using Pearson correlation coefficient (R) and coefficient of determination (R^2).[22,31] R measures the linear relationship between the two distributions. Compared to Mean Absolute Error or Mean Squared Error, R^2 is a more informative goodness-of-fit metric, especially when comparing the performances across the locations with different pollen concentration distributions.[32] Noteworthy, we compute the R and R^2 scores only for days with daily total pollen concentration greater than zero, i.e. focusing on relevant daily total pollen concentrations.

We use the resulting CB models to assess and rank the weight in the past and future contexts of 22 environmental parameters in predicting the daily total pollen concentration for all 23 cities.[23,33] In addition to the term "feature importance", widely used in international literature, in this paper we also use the expression "variable importance" with the same interpretation.[22] We obtain a variable importance score for (i) each of the 22 *past* environmental variables, (ii) each of the 22 *future* variables, (iii) the *past* DOY (day of the year), (iv) the *future* DOY, and (v) the past pollen concentration. The score is always non-negative. Each importance score is computed for each combination of the 23 considered cities and the 14 forecast days.

We identify prediction dynamics for each city and forecast day. For a city c (resp. a forecast day f), we compare the distribution of variable importance of c (resp. f) with those of all cities (resp. all forecast days) and detect significant differences by applying a two-sample Kolmogorov–Smirnov test. This results in a characterization of the variable contribution to the different contexts.

Finally, we applied the Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction algorithm[34] in combination with the DBSCAN clustering algorithm[35] to identify similar predictive dynamics across cities and forecast days. The UMAP projected the feature importance vectors (i.e. the weights of the 22 environmental variables, and the DOY, as vectors) of each city-day pair into a two-dimensional space, where the distances reflect the distances in the original space. We use UMAP as a pre-processing step for DBSCAN and to explore the topology of the feature importance dataset. DBSCAN is an algorithm for

TABLE 1 Pearson R and R^2 metrics are presented for the CatBoost (CB) and deep learning (DL) models. The numbers in the model refer to the number of past days used as input, while np indicates that the daily total pollen concentration was not used as an input variable. The Pearson's R p-values are all significant at .001 level.

Model performance parameters	CB_7	CB_7np	CB_14	CB_28	DL_7	DL_7np	DL_14	DL_28	PERS[c]
Pearson R[a]	.712	.639	.712	.714	.716	.650	.705	.702	.469
R^2[a]	.486	.372	.485	.490	.495	.403	.478	.463	−.063
Pearson R[b]	.687	.607	.687	.690	.694	.619	.680	.678	.430
R^2[b]	.449	.326	.448	.452	.462	.361	.439	.424	−.137

Note: CB_7np and DL_7np: models without the past daily total pollen concentration. The values in the columns of CB_7 and DL_7 in Table 1 do not match the values in the last two rows of Table S4. The reason is that in Table 1 we show the average of the model performances over the 14 future days, while in Table S4 we show the metrics for specific forecast days.

[a]The metrics are computed considering all (including 0) daily total pollen concentrations.

[b]The metrics are computed considering only the daily total pollen concentrations greater than 0.

[c]Persistence model;

identifying clusters of arbitrary shape.[35] We use it to group city-day pairs with similar variable importance scores (see Supplement, Section S2.2).

3 | RESULTS

To establish a general model performance comparison, we first show the experimental results as an average of the results obtained over the forecast days and cities, to show the global performances of the models (Table 1). The persistence model (PERS) is outperformed by all models, either with or without the past daily total pollen concentration. We report performance metrics computed on the whole test dataset as usually done in the literature (Table 1), considering only daily total pollen concentrations greater than 0 (Table 1). In the following, we emphasize model results on days with a pollen concentration greater than 0 (57.76% of the total dataset).

Using past pollen information as an input feature improves the forecasting performance of CatBoost (CB_7) and deep learning

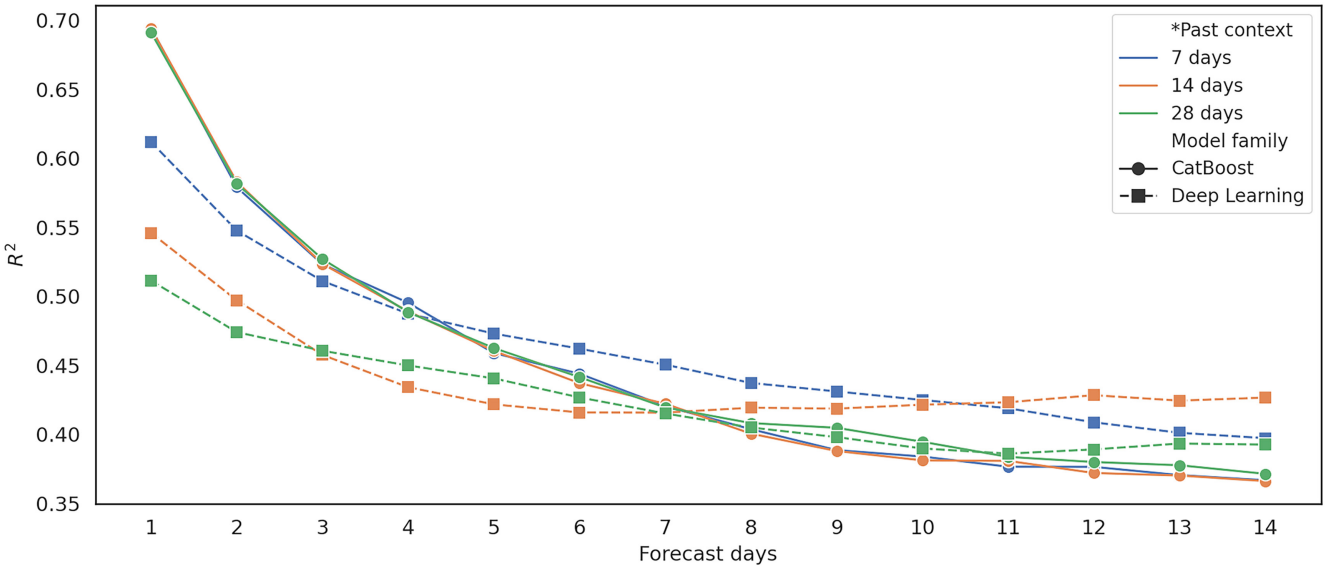


FIGURE 2 Modelled R^2 for each day, averaged over the 23 cities. (The R^2 metrics are computed using only the daily total pollen concentrations greater than 0). (*Based on the environmental variables, including the daily total pollen concentrations of the past 7, 14, and 28 days.)

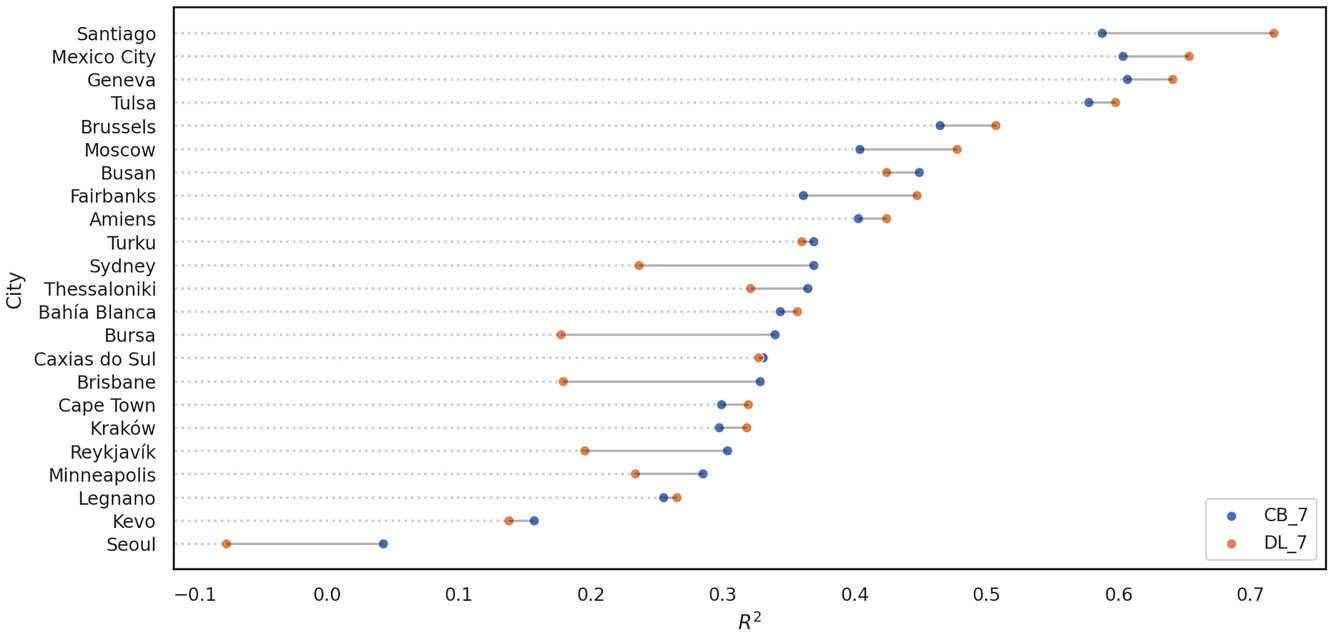


FIGURE 3 Differences in the model performances (R^2 for each city averaged over days) for the best CB and DL models. DL_7 (orange) outperforms CB_7 (blue) for 14 cities. Brisbane (Australia) and Bursa (Turkey) present the largest differences in the performances (The R^2 metrics differences are computed using only daily total pollen concentrations greater than 0).

(DL_7) models compared to models without past pollen information (CB_7np and DL_7np) (Table 1).

Both the CB and the DL models show that increasing the past temporal context over 7 days improves only marginally the forecasting results (Table 1; see also for models R^2: Figure 2). The CB and DL model classes are comparable, although the DL model class is associated with slightly higher R^2 (best R^2: .462 for DL_7) Pearson R (best R: .694 for DL_7).

Since increasing the past input context improves the predictive performance only marginally, we focused on models using $p = 7$ past input days (i.e. CB_7 and DL_7), and therefore a lower number of parameters.

Analysing the daily averaged performances, the CB_7 models performed better when forecasting in the range of 1–3 days [R^2(CB_7) ≈ .69, R^2(DL_7) ≈ .61 for prediction on the 1st day], and show similar performances for Day 4 [R^2(CB_7) ≈ .50, R^2(DL_7) ≈ .49 for prediction on the 4th day], while the DL_7 model obtains better performances for Days 5–14 [R^2(CB_7) ≈ .37, R^2(DL_7) ≈ .40 for prediction on the 14th day] (Figure 2).

The CB and DL forecasting performance shows different behaviour depending on the city (Figure 3). Interestingly, the DL_7 model outperforms the CB_7 model in 14 out of 23 cities in terms of R^2. The greatest differences in the model performance data sets can be observed in decreasing order for Bursa (Turkey), Brisbane (Australia), Sydney (Australia), Santiago (Chile), Reykjavík (Iceland), and Seoul (South Korea) (Figure 3). At the same time, where the performance difference was very large, in five out of these six cities, the DL_7 model performed very poorly compared to the CB_7 model. This may reflect that the DL model suffers more in cities with fewer data available for training or sudden temporal changes in pollen concentration, leading to substantial differences in the performance of the two models in inter-city comparisons (Figure 3). However, when the predictive performances of the two models are similar for a specific city, the output time series closely follows the same trend and differs mostly for daily values (Figure 4). Notably, we did not find any correlation between the number of years available for each station and the model performances of each city. (Additional results are reported in Supplement, Section S3.1).

We used the CB_7 model to rank the environmental factors that affect the pollen concentration forecast for cities and days (see Supplement, Section S3.2). The five most important variables in determining the future daily total pollen concentrations are the past daily total pollen concentration, the future 2 m temperature (i.e. the future temperature at 2 m height above the surface), the past 2 m temperature, the past soil temperature level 3 (namely, the soil temperature 28–100 cm below the surface), and the past soil temperature level 1 (0–7 cm below the surface) (Figure 5; Supplement, Table S2; Supplement II).

The next step consisted of detecting similarities between the cities and model dynamics over the forecast days. From the application of the UMAP (parameters: n_neighbors = 7, min_dist = 0.05, n_dim = 2) and DBSCAN algorithms (parameters: eps = 0.4, min_samples = 4) to the projected samples, we obtained 20 clusters. Each cluster grouped together similar variable importance vectors associated with cities and forecast days (Figure 6A). In addition,

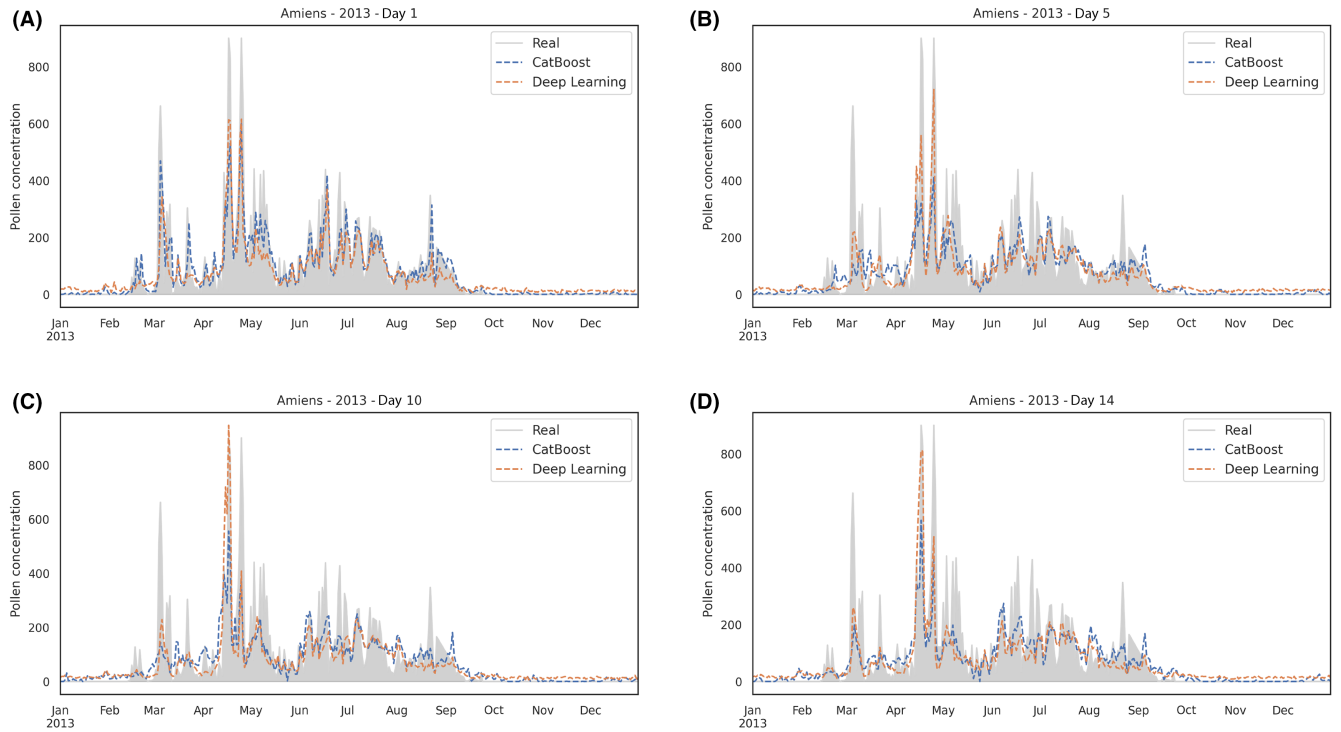


FIGURE 4 Predictions of CB_7 (blue dashed line) and DL_7 (orange dashed line) models for Amiens in the year 2013. The grey background represents the observed pollen count. The forecasts are presented for 1, 5, 10, and 14 days in advance. (A) Forecast for 1 day in advance. (B) Forecast for 5 days in advance. (C) Forecast for 10 days in advance. (D) Forecast for 14 days in advance.

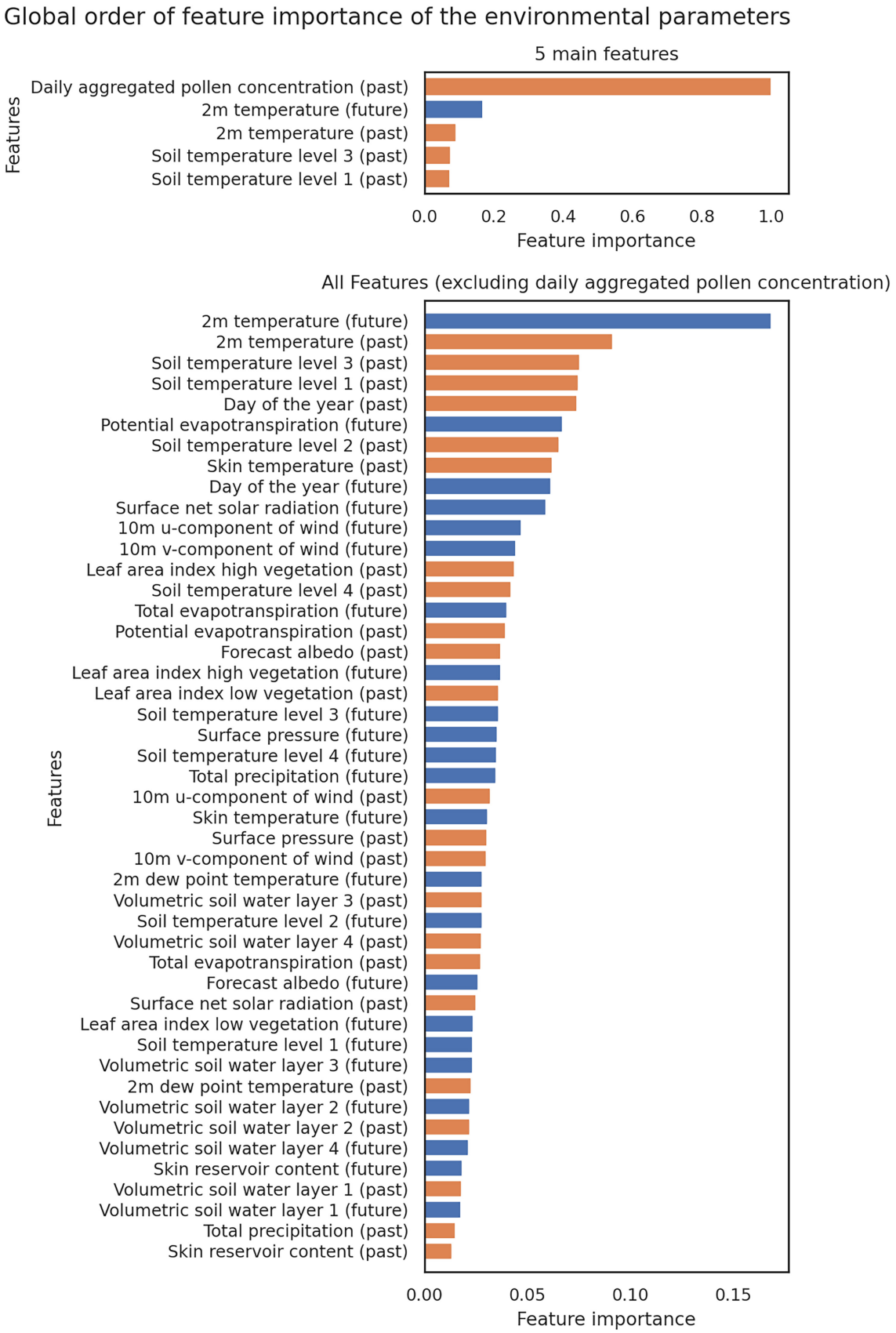


FIGURE 5 Global order of feature importance of the environmental variables resulting from the CB_7 model (Orange: Values of the past variables importance; Blue: Values of the future variables importance). The order of feature importance is averaged over cities and forecast days to describe variables effect on the daily total pollen concentration forecasts. Top: Only the five most important (or top-ranked) variables are shown. Bottom: All variables are reported excluding the daily total pollen concentration variable, for visualization purposes (Note that pollen importance is set to 1, and all other variables importance are given relative to the pollen importance, e.g., 0.2 means as important as one-fifth).

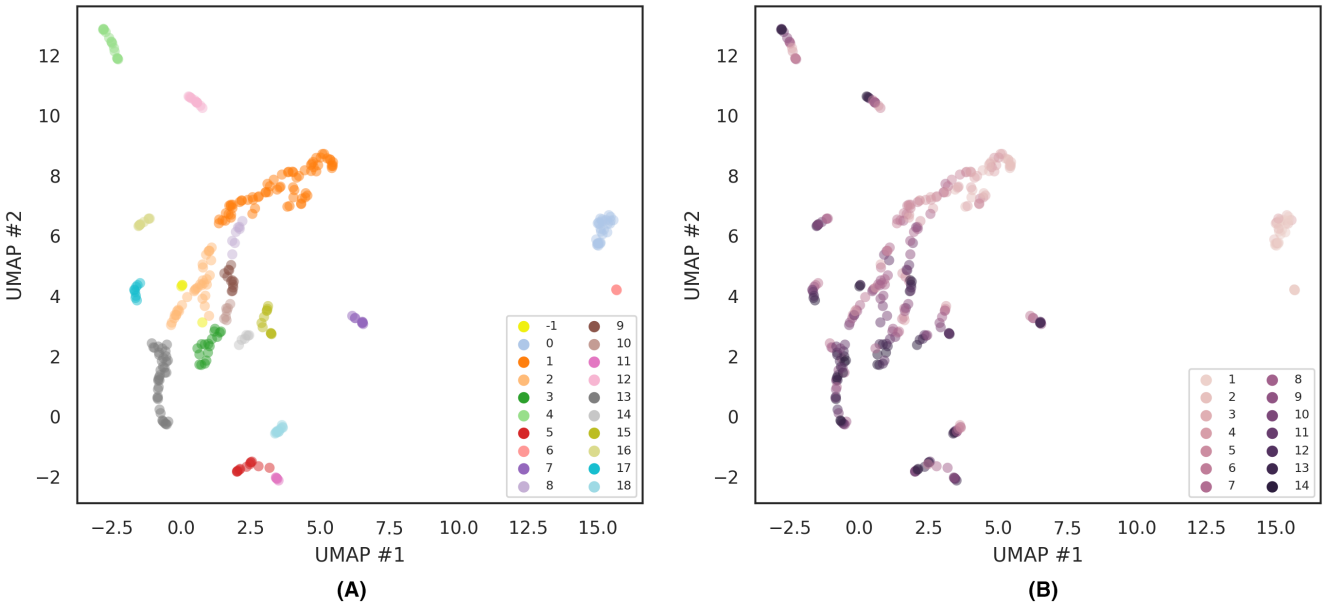


FIGURE 6 The variable importance vectors of the values of the environmental variables have been projected to two-dimensional vectors with UMAP and clustered with DBSCAN. Each data point represents one of the 322 pairs of 23 cities and 14 forecast days. (A) Each cluster is represented with different colours and is composed of at least 4 data points. Noisy points (i.e., those that do not belong to any cluster) are associated with the value −1. (B) Each data point is coloured based on the future prediction day (see Figure 7). In Figure 6A, in the center-right area, there are city-day pairs (greyish-blue) associated with low values of forecast days (see Figure 7, cluster 0) (i.e., similar features tend to be useful for the 1st forecast days across different cities).

each data point is coloured based on the future prediction day (Figure 6B). For example, cluster 0 indicates that the same environmental variables are important for 17 cities out of 23 for predicting the daily total pollen concentration for the 1st forecast day, and in some cases also for the following days (Figure 7). Additionally, the same features that are useful for predicting the daily total pollen concentration for the 1st forecast day, are also relevant for predicting pollen concentration for the second forecast day at Cape Town (South Africa), Brisbane (Australia), Mexico City (Mexico), Bursa (Turkey), Thessaloniki (Greece), and Santiago (Chile) (Figure 7; Supplement, Section S3.3). Cluster composition was characterized by comparing the distribution of feature importance values of the environmental variables of all cities and forecast days in a given cluster with the distribution of feature importance values of all other cities and forecast days. For example, we compare the distribution of the environmental variable importance of the elements of cluster 8 (there are a total of 8 elements from cluster 8; see Figure 7) with the distribution of the environmental variable importance of all other cities and forecast days (23 cities x 14 forecast days = 322 elements–8 elements = 314 elements; see Figure 7). A two-sample Kolmogorov–Smirnov test was conducted to identify only the significant variables importance for each cluster, selecting the environmental variables associated with a $p < .005$, or .5%. For example, for city-day pairs included in cluster 0, model predictions are heavily based on the past pollen concentration values and are less sensitive to for example the forecasted temperature (i.e. 2 m temperature (future)) (Figure 8, dark boxplots), compared to the distribution of its feature importance in other clusters (Figure 8, light boxplots), global order of feature importance of the environmental variables (Figure 5; see Supplement, Section S3.4).

4 | DISCUSSION

Considering that, according to the World Allergy Organisation, pollen allergies are a global public health issue,[36] confirmed by the effect of climate change on increasing pollen loads and trends,[37,38] the development and improvement of airborne pollen forecasting models are crucial to prevent pollen allergy sufferers from being exposed to high levels of aeroallergens in the near future.

Our study is a completely unique work on the subject, because (1) It uses two up-to-date AI methods to forecast the pollen concentration 14 days in advance, and also compares their goodness; (2) It is the first in the international literature to forecast the total pollen concentration for 23 aerobiological station of all five inhabited continents, for which it uses the most comprehensive pollen—environmental parameter database to study the pollen-climate relationship system; (3) The model includes the projected environmental parameters, recent total pollen concentrations (1, 2 and 4 weeks), the past environmental explanatory variables and, as a novelty, their future values; (4) Another innovation of the study is that the rank of importance of the environmental parameters used in the study for calculating of the total daily pollen concentration were determined per station and then averaged over the 23 stations in decreasing order; (5) A further novelty is the clustering

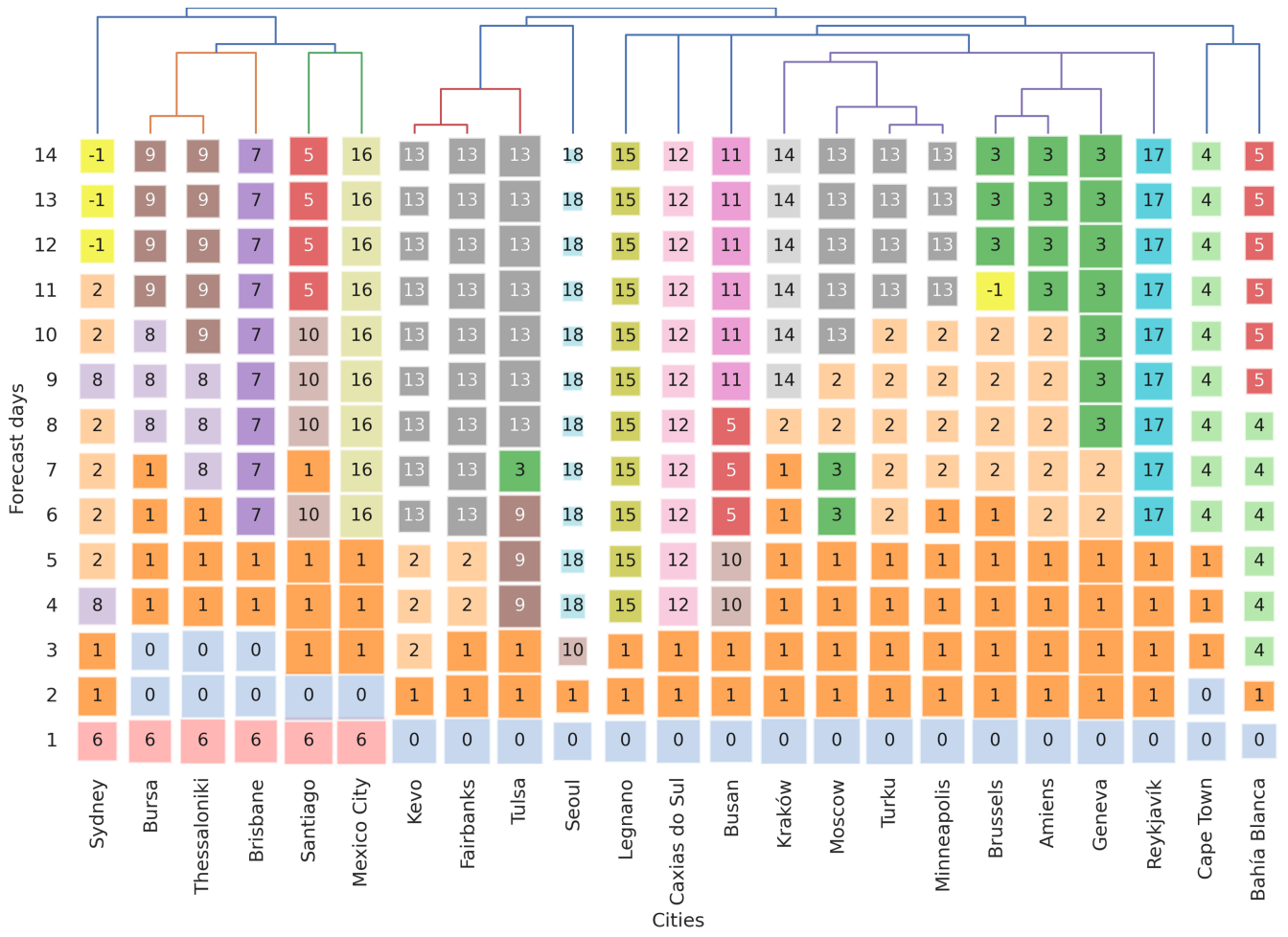


FIGURE 7 Clustering results for all combinations of city-day. The numbers in the squares are the serial numbers of the 19 clusters from 0 to 18, comprising groups of various environmental variables. A value of −1 indicates points classified as noise from the DBSCAN algorithm (points that are not part of a group of at least 4 points). For example, cluster 0 indicates that the same features (i.e., the same environmental variables) are important for 17 cities for predicting the daily total pollen concentration for Day 1, i.e., for the next day. At the same time, for Day 2 the same environmental variables in cluster 0 are important only for six cities. Larger squares indicate a higher predictive performance for the city-day combination (smallest square: Seoul (South Korea), day 12, $R^2 = 0$; biggest square: Mexico City (Mexico), Day 1, $R^2 = 1$; median size square: Busan (South Korea), Day 7, $R^2 = .506$). The dendrogram at the top indicates cities that are similar in terms of the clusters they are assigned to.

of the joint distribution of the daily feature importance values of the environmental variables according to cities, which reveals their possible changes on consecutive forecast days; (6) Another novelty is that, for the first time in the international literature, we performed a prediction of the total daily pollen concentration per station, taking into account both allergenic and non-allergenic pollens, through which the combined frequency of allergic and non-allergic rhinitis can be predicted.[39,40]

Overall, this new knowledge of the ecological relationships of the most remarkable variables importance for pollen forecast models according to clusters, cities and forecast days is important for developing and improving the accuracy of airborne total pollen forecasts.

We studied the daily total pollen concentration, which is the sum of the daily pollen concentration of all pollen-dispersing taxa. This is a novel approach in the pollen-related literature, as most papers focus on the environmental relationship of a single species.

We apply CB and DL models to estimate the daily total pollen concentration. Random forest and DL in Zewdie et al.[23] have reached a Pearson *R* of .82 and .81. The CB_7 and DL_7 models scored 0.83 and 0.82, training on the same dataset. A neural network developed for predicting *Parietaria* pollen concentration for the next 3 days has obtained an average R^2 of .50, in the Atlantic coastal regions of Spain.[19] The CB_7 and DL_7 models scored 0.62 and 0.59 (globally averaged R^2 for the next 3 days forecast). Lops et al.[31] have developed a convolutional neural network that obtained Pearson *R* of .83 and .69 for 1st and 7th day predictions. Our CB_7 and DL_7 models scored .85 and .82 for the 1st day forecast, and 0.69 and 0.71 for the 7th day forecast. Overall, the performance of our model predictions is among the best compared

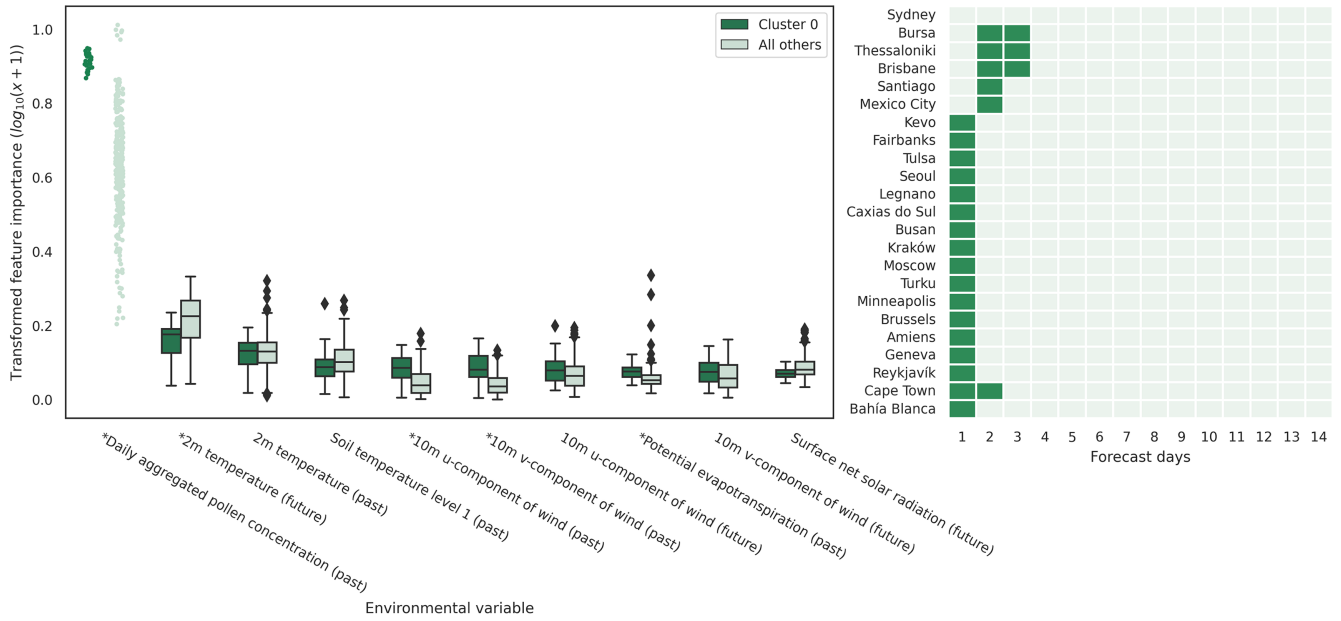


FIGURE 8 The top 10 most important environmental variables in decreasing order of the median of their feature importance. Values from left to right are for predicting the daily total pollen concentrations, cluster 0, city-day combinations. (For better clarity, instead of all 47, only the first 10 most important environmental variables are shown.) Feature importance values were transformed using the log10 (x + 1) transformation for visualization purposes. Feature importance values of the daily total pollen concentration (past) both for cluster 0 and all other clusters are presented with a strip plot to highlight the number of points belonging to each group. The diagram on the right indicates what city and forecast day pairs are included in the cluster. *Significant differences between cluster 0 ($p < .005$) and all other clusters.

to other models (more extensive performance comparisons with other models in the literature can be found in Supplement, Section S4.1).

Globally averaged R^2 scores indicate that the models do not explain more than 50% of pollen concentration variability. Although the predicted pollen concentration values could derive from the actual observations, the models forecast with reasonable accuracy the dynamic, forecasting the occurrence of peaks, which are of great concern for pollen allergy sufferers.

By ranking the importance of input features for the CB model it becomes clear what variables are the most important in a context of limited data or resource availability. However, the resulting feature ranking is highly dependent on the whole experimental setting (i.e., modelling technique, available data set, and considered environmental conditions). In addition, the use of UMAP and DBSCAN allows to suggest what prediction dynamics are common among different scenarios (e.g., climates and future days). For example, cluster 0 indicates that past pollen concentration and past and future wind intensities are the most important features for predicting 2nd and 3rd day pollen concentration in Bursa (Turkey), Thessaloniki (Greece), Brisbane (Australia), Santiago (Chile), and Mexico City (Mexico) (Figure 8). Such a result can help highlight what aspects contribute the most to pollen concentrations of various areas. The same tool could also be used to compare more delimited areas (e.g. regions or states).

Zewdie et al.[25] used the ERA5-Land reanalysis data set to produce predictor parameters for their machine-learning pollen estimates. Only 13 out of the 22 environmental variables used in our study are common to Zewdie et al.[23] (Supplement, Table S2). Therefore, it is difficult to compare the conclusions of the two papers in the prediction of the pollen concentration of the environmental parameters, as well as their order of importance established therein. Another similarity is that their studied station, i.e. Tulsa (OK, USA),[23,41] is one of the 23 studied stations in our paper.

On the whole, the performance of our model predictions is among the best compared to other models (Supplement, Table S4).

The role of past pollen is the most prominent in shaping future pollen concentration, compared to past or pollen variables. For some cities [Brisbane (Australia), Bursa (Turkey), Mexico City (Mexico), Sydney (Australia) and Thessaloniki (Greece)] the past pollen variable plays a particularly important role (Supplement, Figure S8), which is consistent with previous forecast evaluation for Brisbane and Sydney (both Australia), where the importance of persistence of pollen concentrations was also found to be major predictor.[42]

The five most important variables in determining the future daily total pollen concentrations mentioned in Section 3 (Figure 5; Supplement, Table S2; Supplement II) differ from those identified in Zewdie et al.[23]; however, it is probably due to the different input datasets and modelling techniques (see Supplement, Section S4.1.1).

The daily total pollen concentrations of the past days play by far the most prominent role in determining the future daily pollen concentrations for the CB_7 model. This can be explained by the fact that during the flowering phenophase, the pollen release of the

previous days is a decisive indicator of the pollen release of the following days. Besides, wind, pollen deposition, rainfall, and air humidity have the opposite effects on the past pollen concentration. In Brisbane and Sydney (both Australia), the importance of these factors may increase due to the longer pollen season because of the meteorological conditions regulating source land vegetative greenness, and the extended or delayed flowering of the different grass species, which constitute up to 70% of total seasonal airborne pollen in the subtropical site of Brisbane (Australia) (Figure 5; Supplement, Figures S9 and S10).[43] Further discussion of the role and ranking of relevant environmental features can be found in the Supplement, Section S4.2.

5 | CONCLUSIONS

The development of these models allows airborne pollen prediction on a global scale that can play a crucial role in the daily management of pollen allergy symptoms and could have a significant impact on the quality of life of allergy sufferers worldwide.

Using the past pollen information improves the forecasting performances over both the CB and DL models. The CB_7, CB_14, CB_28, and DL_7 were found the best, whereas the DL_28 model provided the weakest forecasts. The best pollen forecasts were made for Santiago (Chile) (R^2(DL_14) ≈ .77), Geneva (Switzerland) (R^2(DL_7) ≈ .67), Mexico City (Mexico) (R^2(DL_7) ≈ .65), and Tulsa (OK, USA) (R^2(DL_28) ≈ .64), while the weakest forecasts were received for Bahía Blanca (Argentina) (R^2(CB_14) ≈ .23), and Seoul (South Korea) (R^2(CB_28) ≈ .12) (R^2 computed on the future 14 days forecasts).

City-related clusters of the most similar distribution of feature importance values of the environmental variables do not change or only slightly change on consecutive forecast days for Caxias do Sul (Brazil), Cape Town (South Africa), Brisbane (Australia), and Mexico City (Mexico), while the clusters often change from one forecast day to the other for Reykjavík (Iceland), Turku (Finland), and Minneapolis (MN, USA). Ecological relationships of the most remarkable variables importance according to clusters, cities and forecast days are also presented.

For short-time forecasts (up to 4 days) the CB model is generally the best choice, while for longer periods the DL model is more accurate. Specific locations showed different performance trends; thus, a comparison should be made where possible. The number of days for which a forecast can be considered reliable depends on two additional aspects: the acceptable error rate for a specific application and location, and the reliability of meteorological forecasts.

First in the literature, not only the past feature importance (weights of the environmental variables) but also future variables importance is determined when calculating the global order of feature importance of the environmental variables resulting from the CB_7 model. Overall, the five most important environmental variables in determining the daily total pollen concentration, in decreasing order, are the past daily total pollen concentration, future 2 m temperature, past 2 m temperature, past soil temperature in 28–100 cm depth, and past soil temperature in 0–7 cm depth. This is the first paper in the literature presenting a comparative analysis for (1) total daily pollen concentration forecasts, (2) the importance of environmental explanatory variables, and (3) the ecology of variables importance for a larger group of cities, covering all five continents, on a global scale.

AUTHOR CONTRIBUTIONS

L.M., L.C., A.G. designed the research; L.M., L.C., A.G., G.J., C.F., M.B., L.H.Z., J.J.H., A.D., G.T., Á.J.D., J.P., C.H.K., P.J.B., M.P.P.G., C.T.-H. performed research; L.M., L.C., A.G., contributed analytic tools; L.M., L.C., A.G., G.J., C.F., M.B. analyzed data; L.M., L.C., A.G., L.H.Z., A.D., D.V., A.C., P.J.B., J.M.D., M.P.P.G., C.T.-H. wrote the paper; L.M. coordinated the study; I.I., L.Cz., E.M., Z.D., C.O. performed data preparation; S.K.H., N.B., A.P., A.S., L.T., M.L-K., S.P., M.T., G.O., A.D., A.C., D.V., S.H., E.R.G, M.B., S.C., J-W.O., K.S., L.F., M.K., E.L., D.M., E.S., R.G., M.C.C-E., C.G.G., M.A.L-G., G.D.R., L.B.B., D.B., J.M.D., P.J.B., S.M.V., R.M.V-B. provided airborne pollen data.

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CONFLICT OF INTEREST STATEMENT

The authors have declared that they have no potential conflict of interest.

DATA AVAILABILITY STATEMENT

Data will be made available from data owners per station upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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